

HARP: ExactFrac-Constrained Score Serving for GNN Prediction APIs

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Abstract—Graph neural network (GNN) prediction APIs power ranking, routing, and analytics on graphs with thousands to millions of nodes—but releasing full confidence scores can leak whether a node was in the training set. Standard privacy fixes add noise to *every* response: that hides membership cues, yet it also destroys the raw model scores that downstream systems need for calibration, threshold decisions, and compliance. Caching or reseeding that noise can make answers *repeatable*, but it still does not return the model’s true posterior. We formalize this missing service contract as **ExactFrac**—the fraction of API responses that equal the unmodified model output—and ask how to maximize accuracy subject to $\text{ExactFrac} \geq c$ (study default $c=0.60$). **HARP** answers that question with selective score serving: it ranks risky nodes, expands a hop neighborhood, protects only that set, and leaves a configurable clean majority exact. Under defense-aware LiRA audits ($n_{\text{shadows}}=16$) and large-graph serving probes, **HARP** is feasible whenever $c>0$ —where uniform and cache-only noise are not—while keeping competitive accuracy and calibration; residual risk on the exact slice is measured and disclosed as a structural limit of $\text{ExactFrac}>0$.

Index Terms—big data prediction APIs, graph neural networks, response caching, **ExactFrac**, selective score serving, membership audit

I. INTRODUCTION

A. Motivation

GNN prediction APIs return softmax vectors for triage, ranking, and analytics at Big-Data scale [9], [10], [28]: graphs with 10^3 – 10^6 + nodes, high query velocity (clients re-query the same entities), and veracity requirements for downstream score consumers. Production serving stacks hash requests and cache responses to cut latency under repeated queries [40], [41]. Released scores also leak training membership [1], [2], [4], [5]. This paper is about the *systems* intersection of those facts: how to release scores under a raw-posterior service contract while still spending noise where an audit requires it—not a new membership-inference theory paper.

a) *Why ExactFrac is operationally necessary.*: Two contracts are easy to confuse. *ReplayFrac* asks whether two queries return the same vector (a cache hit or sticky/seeded draw can satisfy this with a *noisy* first answer). *ExactFrac* $\equiv$ *RawExactFrac* asks whether the released vector equals the unmodified model posterior $\hat{p}_v$. Raw equality matters when downstream systems treat API scores as model truth:

temperature calibration, threshold routers that must match offline evaluation, A/B analytics that join on score bits, and compliance pipelines that attest “this is the model’s output.” Caching a noisy draw restores *ReplayFrac* and *QPS*; it does *not* restore *ExactFrac*. We therefore study

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{ExactFrac}(\Pi) \geq c, \quad \text{LiRA}(\Pi) \leq \tau$$

(optional τ), with study default $c=0.60$ as an operator dial (Fig. 2). Fresh uniform LBP has both fractions ≈ 0 (Remark 1). Table VIII shows global-cache and seeded LBP achieve *ReplayFrac*=1 with *ExactFrac*=0; selective release (**HARP**) meets $c>0$ without a ledger of non-raw scores.

b) *How to read Acc / ExactFrac / LiRA.*: **Acc** is task utility (primary objective under the SLA). **ExactFrac** is the service constraint: fraction of nodes released as raw $\hat{p}_v$. **LiRA** is an optional audit dial—used to decide how much *Frac* to spend, not the headline win condition. In practice: if $\text{ExactFrac} \geq c>0$ is required, discard rows with *ExactFrac*=0 even if their *LiRA* looks better; if only *Acc* s.t. $\text{LiRA} \leq \tau$ matters, prefer equal-mass LBP and ignore **HARP**. Our running Cora example ($c=0.60$, $n_{\text{sh}}=16$): equal-mass LBP $\text{Acc} \approx 0.87 / \text{LiRA} \approx 0.59 / \text{ExactFrac} = 0$ vs. **HARP** $\text{Acc} \approx 0.83 / \text{LiRA} \approx 0.67 / \text{ExactFrac} = 0.60$ with much lower clean-majority ECE.

c) *Framework.*: **HARP** is a constrained score-serving layer for GNN APIs: (1) a *constructor* ranks nodes (topology, degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability); (2) hop expansion $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$ plus a *protector* only on $\mathcal{P}_k$; (3) operator dials *Frac*/Mass/session budget B , with **CFS** searching *Frac* under (c, τ) . Default: *release-only* selective Laplace at *Frac*=0.40 (*ExactFrac*=0.60).

d) *Headline evidence.*: At $n_{\text{shadows}}=16$, **HARP** meets $\text{ExactFrac} \geq 0.60$ while uniform and stateful-noisy LBP cannot; equal-mass LBP remains the right comparator when $c=0$ is allowed. Within the constrained class, constructors cut clean-slice AUROC ($0.669 \rightarrow 0.594$), **CFS** selects *Frac* under operator (c, τ) , and **HARP** $\circ$ **GAP** preserves **GAP**’s ε for public-metadata constructors. ogbn-arxiv/products probes show *Acc* and serving cost at volume; privacy audits at that scale are scoped and canary-stressed (not claimed as full-graph *LiRA*).

B. Gap

Prior release defenses (LBP [5], MemGuard [6]) perturb every response and therefore yield $\text{ExactFrac}=0$ unless wrapped in a response ledger that stores non-raw scores. Train-time GNN MIA methods (GTD, MaskArmor) [21], [22] and DP-GNNs [13], [14], [36] do not expose $\text{ExactFrac}/\text{Mass}$ serving accounting for prediction APIs. None formalize Acc under $\text{ExactFrac} \geq c > 0$ as a Big-Data score-serving objective with constructors, validation-time Frac search, and trainer composition.

C. Contributions

- 1) **ExactFrac-constrained API release.** Separate RawExactFrac from ReplayFrac for prediction serving; show fresh and stateful-noisy uniform policies fail $\text{ExactFrac} \geq c > 0$; hop coverage and DP post-processing composition (Prop. 2).
- 2) **HARP serving layer.** Selective, hop-aware release with constructors, validation-time CFS, optional DCS, and $\text{HARP} \circ \text{GAP}$ for finite ε plus ExactFrac controls under continuous query traffic.
- 3) **Constraint-aware Big-Data evaluation.** Headline LiRA at $n_{\text{sh}}=16$; RawExactFrac -vs- ReplayFrac SLA (Table VIII); cache microbenchmarks; scoped OGB $\text{Acc}/\text{serving}$ and canary-stressed probes; clean-slice risk as a structural limit.

II. BACKGROUND AND RELATED WORK

Shadow/confidence MIAs [1]–[3] and LiRA [4] are the audit standard for prediction APIs; overconfidence is a recurring cue [17]. On graphs, Olatunji et al. [5] introduce node-level MIAs and LBP; He et al. [38], GTD [21], MaskArmor [22], and surveys [8] study related attacks/defenses. MemGuard [6] adversarially perturbs every confidence vector ($\text{ExactFrac}=0$). DP-GNNs such as GAP [13], [14] provide train-time (ε, δ) ; structure modulates leakage [18], [39]. **Positioning:** This is a prediction-API / score-serving paper with an optional membership audit dial. LBP/MemGuard spend on every node ($\text{ExactFrac}=0$ unless a non-raw ledger is added); GTD/MaskArmor harden training without ExactFrac accounting; GAP answers formal ε ($\text{ExactFrac}=1$ at eval for train-only noise); HARP answers $\text{ExactFrac} \geq c$ serving under continuous query traffic and composes with DP trainers under Prop. 2. Our GAP-agg and MemGuard baselines are faithful reimplementations, not line-by-line ports of the official codebases—we label them as such throughout. Primary comparisons under $c > 0$ are selective policies at matched Frac ; equal-mass LBP is the unconstrained ($c=0$) reference.

III. PROBLEM SETUP

A. Notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ nodes, feature matrix $X \in \mathbb{R}^{n \times d}$, and labels $y \in \{0, \dots, C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{Q}$ with ratios 40%/20%/40% (resampled per seed). Training is *transductive*: every node participates in message

passing; only $\mathcal{T}$ enters the cross-entropy loss. We write $\hat{p}_v$ for the softmax posterior at v , r_v for a risk score in $[0, 1]$, $\mathcal{P}_k$ for the hop-expanded protected set, $\text{Mass} = \sum_v \sigma_v$, and $\text{Frac} = |\mathcal{P}_k|/n$.

B. Constrained score release

Operators rarely optimize Acc alone. They may need usable accuracy *and* a positive fraction of unmodified model posteriors, subject to a membership audit.

Definition 1 (Constrained Score Release). *Given a trained GNN and release channel, choose a release policy Π to*

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{LiRA}(\Pi) \leq \tau, \quad \text{ExactFrac}(\Pi) \geq c,$$

where $\text{ExactFrac} \equiv \text{RawExactFrac} = \Pr_v[\tilde{p}_v = \hat{p}_v]$ (fraction of nodes released as the unmodified posterior), $\text{ReplayFrac} = \Pr_v[\tilde{p}_v^{(1)} = \tilde{p}_v^{(2)}]$ is reported separately, τ is an operator-chosen audit risk tolerance matched to the LiRA budget (not a universal privacy target), and $c \in [0, 1]$ is an operator-chosen clean fraction (study default 0.60). Under release-only selective Laplace with $\text{weak}=0$, $\text{ExactFrac}=1 - \text{Frac}$ by construction and implies ReplayFrac on the clean slice without a response ledger.

Remark 1 (Uniform and stateful-noisy LBP under $c > 0$). *Independent Laplace noise with scale $b > 0$ on every coordinate changes $\hat{p}_v$ almost surely, so $\text{ExactFrac}=0$ for any uniform fresh LBP policy. A global per-node response cache or a deterministic PRNG keyed by (node, epoch) can raise ReplayFrac to 1 and restore cache hits / ledger audit, but ExactFrac remains 0 (Table VIII). Selective release with protected fraction Frac achieves $\text{ExactFrac}=1 - \text{Frac}$ by construction, without storing non-raw scores.*

This reframes the Acc –LiRA comparison. Equal-mass LBP can win Acc subject to $\text{LiRA} \leq \tau$ when $c=0$ is allowed; HARP solves the strictly harder problem with $\text{ExactFrac} \geq c > 0$. The Frac sweep traces HARP’s Pareto frontier under that constraint.

C. Threat model

The adversary is a black-box client: it queries softmax scores and tries to guess whether a node was in the training set [2], [4], [5]. It does not see gradients or embeddings. White-box parameter access remains out of scope; label-only gap AUROC and edge-similarity membership AUROC are reported alongside LiRA (GAP-agg cuts edge AUROC $0.895 \rightarrow 0.748$ on Cora).

We also stress three practical variants: defense-aware shadows, multi-query averaging ($K \in \{1, 5, 20\}$), and an LTE-aware query adversary that focuses on clean nodes, protected-set boundaries, and top-decile LTE seeds (Sec. VI-B01). We do not claim robustness to poisoning or unbounded queries without a session budget.

HARP: ExactFrac-constrained selective score serving

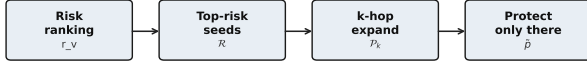


Fig. 1. HARP pipeline: risk ranking $\rightarrow$ high-risk seeds $\rightarrow k$ -hop expansion $\rightarrow$ protector only on the protected set (ExactFrac=1 - Frac on the clean majority).

D. Metrics

Utility: test Acc (primary). **Audit:** LiRA AUROC [4] and TPR@1% FPR, plus Salem-style ϕ /confidence attacks [2]. **Fidelity:** ECE [17], clean-slice ECE, ExactFrac. LiRA decides *whether* to spend noise; Acc/ECE measure the cost.

Remark 2. *The primary protocol is transductive (non-members remain in the graph during training), matching common semi-supervised GNN APIs. We additionally report an inductive edge-exclusion check on Cora (Sec. VI-B0m) where test-incident edges are dropped at train/infer time.*

IV. HARP: HOP-AWARE SELECTIVE RELEASE

A. Design principle

Uniform *fresh* posterior noise cannot meet ExactFrac $\geq c > 0$ —any independent $b > 0$ draw changes $\hat{p}_v$. Stateful wrappers (global caches, ledgers, seeded PRNGs) can meet ReplayFrac without ExactFrac (Table VIII). Node-local selection without hop coverage is incomplete for GNNs: if v is protected but neighbor u is released exact, aggregation can reintroduce membership signal into u 's score. HARP is a three-interface serving framework (Fig. 1):

- 1) **Constructor (pluggable):** rank nodes—topology (LTE), degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability.
- 2) **Geometry + protector:** expand $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$; apply Laplace or masking only on $\mathcal{P}_k$.
- 3) **CFS + dials:** validation-time search of Frac under ExactFrac $\geq c$ and optional LiRA $\leq \tau$; log Mass/Frac/ B .

Locked study config: *release-only* Laplace, Frac=0.40 (ExactFrac=0.60), $k=1$, $\sigma=0.30$. Topology LTE is the no-shadow default constructor; ensemble/audit-guided ranking is preferred when shadows are already paid for. Train-time alignment is an optional ablation—not required for the ExactFrac claim.

a) *Noise mass and clean fraction.*: Mass = $\sum_v \sigma_v$ and Frac = $|\mathcal{P}_k|/n$ are operator accounting metrics. Uniform LBP at scale b has Mass= bn and Frac=1. See paragraph IV-E0a for the locked-budget identity.

b) *Locked-config example (Cora).*: On Cora ($n=2708$), Frac=0.40 protects ≈ 1083 nodes after one-hop expansion; ≈ 1625 nodes stay exact. Mass_{HARP} ≈ 325 matches equal-mass LBP with $b=0.12$. Strong LBP ($b=0.3$) spends Mass ≈ 812 on all nodes—the Acc-collapse baseline in Table II.

c) *Release transform (Laplace protector).*: Let $\sigma_v = \sigma_{\text{strong}} \mathbb{1}[v \in \mathcal{P}_k]$. The released posterior is

$$\tilde{p}_v = \text{renorm}([\hat{p}_v + \text{Lap}(0, \sigma_v)]_+), \quad (1)$$

with independent Laplace coordinates and the same transform on every defense-aware shadow. The masking protector instead returns $\hat{p}_v$ for $v \notin \mathcal{P}_k$ and a one-hot of $\arg \max \hat{p}_v$ for $v \in \mathcal{P}_k$.

d) *Algorithm (systems view).*:

- 1) **Risk:** $r \leftarrow \text{Constructor}(G, \mathcal{T})$ (topology / degree / confidence / random / audit).
- 2) **CFS (optional):** search Frac $\leq 1 - c$ to maximize Acc subject to LiRA $\leq \tau$.
- 3) **Seeds + expand:** binary-search seeds so $|\text{Ball}_k(\mathcal{R})|/n \approx \text{Frac}$; $\mathcal{P}_k \leftarrow \text{Ball}_k(\mathcal{R})$.
- 4) **Release:** apply (1) (or masking); log (Mass, Frac, Acc, LiRA, ECE).

Train-time alignment is optional and omitted in the locked release-only configuration.

B. Risk constructors

The constructor interface is part of the framework contribution. **Topology (LTE)** uses only $G, y|_{\mathcal{T}}$, and the train mask (inverse degree, local heterophily, train-neighbor exposure); it is a cheap default when shadows are unavailable—not claimed to be privacy-optimal. **Degree / train-neighbor** isolate individual topology cues. **Confidence / entropy** rank from a clean forward pass (high max-confidence or high entropy). **Random** is the matched-Frac selectivity control. **Audit-guided** scores $|\mu_{\text{IN}} - \mu_{\text{OUT}}|$ over a small shadow ensemble [37] and is preferred when that ensemble is already paid for. We report Spearman correlation between each constructor and audit vulnerability, plus Acc/LiRA/slice AUROC at matched Frac (Table VII). On Cora/Citeseer, *entropy* is the strongest cheap prior (Spearman ≈ 0.33 – 0.35); topology (LTE) is moderate (≈ 0.25 – 0.28); max-confidence is anti-correlated. Heterophilic Chameleon priors are weak—selectivity still enforces ExactFrac, but ranking gains are limited. Where topology underperforms random on LiRA (e.g., Cora in Table VII), we state that plainly: ExactFrac comes from selectivity, not from LTE.

C. Why k -hop expansion

Let $\mathcal{R}$ be seeds. A GNN score at u already mixes information from $\text{Ball}_k(u)$ before any release noise is applied. If a membership-sensitive seed $v \in \mathcal{R}$ is noised at release but a neighbor u is returned bit-exact, the adversary simply queries u .

Proposition 1 (Hop coverage under a one-hop linear mixer). *Fix a one-hop score model $s_u = \alpha z_v + (1 - \alpha) z_u$ for $u \in \mathcal{N}(v)$, $\alpha \in (0, 1]$, where z_v carries a membership cue of gap $\gamma > 0$, and release adds independent Laplace noise only to chosen nodes' scores. If the policy noises v but leaves u exact, an adversary observing s_u retains the unprotected leaf advantage. Noising $\text{Ball}_1(\{v\})$ places Laplace noise on every score that linearly depends on z_v under this mixer.*

Proof sketch. Under seed-only protection the released leaf equals the clean mixer output, so the member/non-member gap in $\mathbb{E}[s_u]$ remains $\alpha\gamma$. Expanding protection to u subjects that coordinate to $\text{Lap}(0, \sigma)$. A synthetic mixer confirms the ordering: unprotected and seed-only leaf AUROCs stay near 1.0, while hop-1 protection lowers the leaf attack to ≈ 0.72 . $\square$

This is a motivating linearized argument for hop expansion, not a necessity theorem for nonlinear multi-layer GNNs. Empirically we set k to the model’s receptive field (default $k=1$ for 2-layer SAGE/GCN; $k \in \{0, 2\}$ changes Acc by $\lesssim 0.01$ at matched Frac). Frac—not k —is the population Acc–Mass knob.

D. Guarantees: CFS configuration and DP composition

Remark 3 (CFS is validation-time configuration, not a novel optimum). *Constrained Frac Search (CFS) enumerates a finite grid $\mathcal{F} \subseteq [0, 1 - c]$ and returns $f^* = \arg \max\{\text{Acc}(f_i) : \text{LiRA}(f_i) \leq \tau\}$ using validation Acc and a validation-time defense-aware LiRA estimate; the locked tables then report a held-out test Acc / headline LiRA once. This is a reproducible operator procedure for choosing Frac under explicit (c, τ) —not a claim of global optimality over all release policies or constructors. The audit cap τ is an operator risk tolerance tied to the attack budget (study headline: $n_{\text{sh}}=16$, $\tau=0.70$); $\tau=0.55$ is infeasible at that budget, and we report neighboring caps $\{0.68, 0.70, 0.75\}$ (Sec. VI-B).*

Proposition 2 (Post-processing composition). *Let T be any training procedure whose released model satisfies (ε, δ) -DP w.r.t. the training data, and let Π be a HARP release policy that is a (possibly randomized) function only of the model’s released outputs and public deployment metadata. Then $\text{HARP} \circ T$ satisfies the same (ε, δ) -DP (standard post-processing [27]) and attains $\text{ExactFrac}=1 - \text{Frac}$ w.r.t. the DP model’s posteriors.*

Proof. Immediate from the post-processing property whenever Π does not re-read private training labels or private membership bits beyond what T already released. $\square$

Prop. 2 is not a new DP theorem; it is the reason HARP can sit *above* a DP trainer as a serving layer. **Constructor caveat:** topology LTE uses the train mask and $y|_{\mathcal{T}}$ as server-side deployment metadata. If those bits are private under the DP threat model, use a public-metadata constructor (degree, confidence/entropy from the released DP outputs, random, or audit scores computed only from released shadow outputs) before claiming Prop. 2. Section VI-B measures $\text{HARP} \circ \text{GAP}$ under the public-metadata reading.

E. Complexity and accounting

Proposition 3 (HARP setup and release complexity). *Let $G = (V, E)$ have $n = |V|$ nodes and $m = |E|$ adjacency endpoints, with C classes. Fix hop radius $k = O(1)$ and $T = O(1)$ binary-search iterations for a target protected fraction. Then setup (LTE + seeds + k -hop expand) runs in $O(m + n \log n)$*

time and $O(n + m)$ space. Release runs in $O(nC)$ —matching uniform LBP’s tier, with less noise-sampling work whenever $\text{Frac} < 1$.

Proof sketch. LTE scans edges then normalizes n scores: $O(m + n)$. Seed selection is $O(n \log n)$ plus T BFS expansions at $O(m + n)$. Release writes scales, samples Laplace on $\mathcal{P}_k$, and renormalizes: $O(nC)$. $\square$

Remark 4 (Per-response Laplace; no global DP claim). *Interpret each node’s clean posterior as a query with ℓ_1 -sensitivity at most $\Delta=2$ on the simplex. Independent $\text{Lap}(\sigma)$ on coordinates of $v \in \mathcal{P}_k$ is a per-response Laplace mechanism with scale σ ; unprotected nodes receive no noise. Any policy with $\text{ExactFrac} > 0$ has infinite global (ε, δ) -DP by construction. We report this only as an accounting remark—not as a local-DP guarantee under graph adjacency—and recommend training-time DP (e.g., GAP) when a legal global ε is required [13].*

a) *Accounting identity (stated once).*: Under locked Frac and fixed σ_{strong} , $\text{Mass}_{\text{HARP}} = \text{Frac} \cdot n \cdot \sigma_{\text{strong}}$ by construction. Mass is an operational dial correlated with mean ℓ_1 distortion and top-1 flip rate (at matched Mass 325 on Cora: HARP mean ℓ_1 0.349 vs. LBP 0.493); it is not a standard privacy budget. Empirical claims are (i) feasibility under $\text{ExactFrac} \geq c$, (ii) Acc/ECE within the selective class at matched Frac/Mass, and (iii) serving telemetry.

F. What HARP is not

HARP is not differential privacy and is not a claim of lower population LiRA than uniform noise under $\text{ExactFrac}=0$. It is not a claim that randomization “breaks caching”—stateful caches and seeded noise can stabilize non-raw responses. It is a constrained ExactFrac serving framework: spend a protector on a hop-expanded subset, keep a raw-posterior majority, configure Frac with CFS, and audit residual risk—including on unprotected nodes.

V. EXPERIMENTAL PROTOCOL

A. Pre-declared rules

Primary objective: Acc under $\text{ExactFrac} \geq c$ (study default $c=0.60$; swept in Fig. 2). **Primary metrics:** Acc, ExactFrac , ECE (fidelity). **Audit metric:** LiRA AUROC (and clean-slice confidence AUROC) as deployment constraints—not the headline win condition. **Operational metrics:** Mass, Frac, ExactFrac (Raw ExactFrac), ReplayFrac , shared-cache hit rate (Table VIII). Hyperparameters are not selected to minimize test LiRA. Under $c > 0$, primary baselines are selective policies at matched Frac; equal-mass LBP is the unconstrained ($c=0$) reference; sticky/global-cache/seeded LBP are ReplayFrac -positive but ExactFrac -infeasible controls.

B. Systems metrics (definitions)

- **ExactFrac** ($\equiv \text{RawExactFrac}$): fraction of nodes with $\tilde{p}_v = \hat{p}_v$.
- **ReplayFrac**: fraction of nodes with equal responses under re-query (may use cache/seed state).

- **Mass** = $\sum_v \sigma_v$: operational Laplace-scale spend (correlated with distortion).
- **Frac** = $|\mathcal{P}_k|/n$: protected fraction (ExactFrac=1 – Frac under weak=0).
- **ECE**: calibration of released probabilities.

C. Locked HARP (release-only)

Release-only selective Laplace: $\lambda=0$, no HCAG, no train-on-protected; $k=1$, $\sigma_{\text{strong}}=0.30$, Frac=0.40 (ExactFrac=0.60). Constructor default in locked tables: topology (LTE); audit/random/degree/confidence appear in Table VII. Train-time alignment is reported only as an ablation. Baselines: equal-mass LBP $b=0.12$; strong LBP $b=0.3$; MemGuard; GTD; MaskArmor; selective masking; DP-SGD Acc-tuned (vacuous ε).

D. Baselines and attacks

Gaussian LiRA with defense-aware shadows: $n_{\text{shadows}}=16$ for the headline Cora table and CFS (Tables II, V; $\tau=0.70$), $n_{\text{shadows}}=4$ for multi-dataset grids and Acc dials, and sweeps $\{4, 16, 64\}$ on Cora/Chameleon. Five seeds with mean $\pm 95\%$ CI on primary Cora/Chameleon/Citeseer constructor tables; paired seeds across defenses. ogbn-arxiv: NeighborLoader serving Acc with LiRA at $n_{\text{shadows}} \in \{2, 4, 16\}$ (audit stress, not a million-node privacy claim). ogbn-products: full-graph serving RSS/QPS; BFS LiRA ladder at $n \in \{15k, 40k\}$, $n_{\text{sh}} \in \{2, 4, 8\}$, plus a canary-stressed 15k probe with undefended canary-LiRA ≈ 1.0 .

E. What each baseline tests

Equal-mass LBP ($b=0.12$) matches HARP Mass at Frac=0.40 and is the unconstrained ($c=0$) reference. **Selective constructors** (random/degree/confidence/entropy/audit/topology) isolate ranking quality at matched Frac under the SLA. **Strong LBP** ($b=0.3$) is a secondary published recipe that collapses Acc under full-graph strong noise. **MemGuard** (reimplementation) fits a standardized MLP ϕ -attack on holdout members/non-members and searches an L_1 ball to drive $P(\text{member}) \rightarrow 0.5$ while preserving argmax (ExactFrac=0). Following Aerni et al. [37] we report both ϕ and LiRA; at $n_{\text{sh}}=16$ our port cuts LiRA vs. undefended ($0.813 \rightarrow 0.667$) while remaining ExactFrac-infeasible. **GAP-agg** is a GAP-inspired aggregation-Laplace baseline: noise is train-only, so eval ExactFrac=1 with analytical (ε, δ) (hybrid DP-train + ExactFrac-serve). Not a full GAP codebase reproduction. **Selective masking** is an alternate protector (ExactFrac preserved; often worse LiRA). **HARP-full** (Frac=1) is the unconstrained ablation that recovers strong-LBP Acc–LiRA. **GTD/MaskArmor** are train-time defenses (no release Mass/Frac). **Clip+noise DP-SGD** remains an Acc-tuned vacuous- ε reference.

F. Attack protocol (summary)

We train defense-aware shadow models, release transformed posteriors, and report LiRA AUROC with ϕ -features [2], [4].

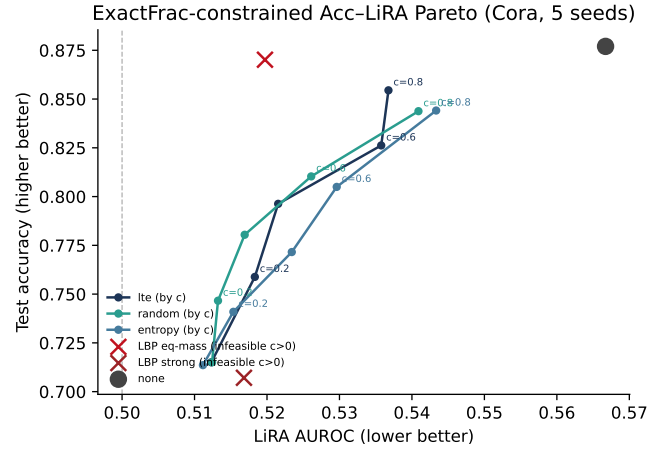


Fig. 2. ExactFrac-constrained Acc–LiRA Pareto on Cora (5 seeds). Selective HARP constructors sweep $c \in \{0.2, 0.4, 0.6, 0.8\}$ (ExactFrac= c). Uniform LBP markers are ExactFrac=0 and infeasible for every $c>0$.

Validation nodes are excluded from membership labels; test Acc/ECE use the held-out test split. Multi-query experiments average $K \in \{1, 5, 20\}$ independent draws; session policies cap fresh draws at $B=5$.

a) *Negative controls.*: PubMed (near-chance undefended LiRA): HARP still preserves Acc vs. strong LBP at Frac=0.40 (0.839 vs. 0.768); ogbn-arxiv/products give near-chance or memory-bounded audits under matched budgets—audit first, harden second [28].

VI. RESULTS

A. Overview

a) *Operational headline (constraint-first).*: Under ExactFrac ≥ 0.60 : (i) HARP is feasible and uniform/stateful-noisy LBP are not (Table I); (ii) release-only HARP keeps Acc competitive within the selective class while returning a 60% raw-posterior majority; (iii) cache microbenchmarks show hit-rate gains when many responses are bit-stable (Table IX); (iv) equal-mass LBP can still win Acc–LiRA when $c=0$ is allowed—that unconstrained privacy-only regime is out of scope for the SLA.

b) *Reading the tables.*: Compare policies only inside the feasible ExactFrac set for your c . Acc is the primary utility number; LiRA/TPR@1% FPR tell the operator whether to raise Frac or tighten τ ; ECE reports score quality on the released (especially clean) majority. Questions: **(Q1)** feasibility under $c>0$; **(Q2)** Acc/ECE vs. selective and equal-mass baselines; **(Q3)** constructors vs. random on utility and slice privacy; **(Q4)** serving cost and clean-slice residual risk; **(Q5)** OGB volume Acc/serving with scoped audits.

B. HARP under ExactFrac ≥ 0.60 : accuracy and score quality

a) *Primary claim.*: Fig. 3, Fig. 4, and Table II: under ExactFrac ≥ 0.60 , release-only HARP is feasible; equal-mass LBP and MemGuard are not. At $n_{\text{sh}}=16$ the undefended attack is strong (LiRA 0.813, TPR@1% 0.200); HARP cuts it to

TABLE I
FEASIBILITY UNDER $\text{EXACTFRAC} \geq c$ (RAW EXACTFRAC).
STICKY/GLOBAL/SEEDED LBP CAN RAISE REPLAYFRAC BUT REMAIN
 EXACTFRAC -INFEASIBLE FOR $c > 0$.

Policy	$c=0$	$c=0.2$	$c=0.6$	$c=0.8$
none	yes	yes	yes	yes
LBP fresh	yes	no	no	no
LBP sticky / global / seeded	yes*	no†	no†	no†
HARP (Frac=1 - c)	yes	yes	yes	yes

*ReplayFrac can be 1 (Table VIII). †ExactFrac=0: responses are never the unmodified $\hat{p}_v$.

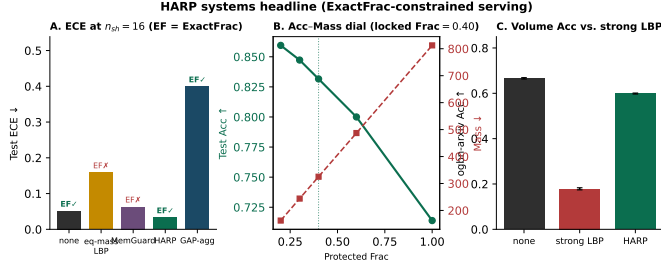


Fig. 3. Systems headline for ExactFrac-constrained serving. (A) ECE at $n_{sh}=16$ with ExactFrac feasibility marks (✓/✗). (B) Frac Acc-Mass dial (locked Frac=0.40). (C) ogbn-arxiv Acc vs. strong LBP (volume Acc, not a full-graph privacy claim).

TABLE II
CORA GRAPHSAGE UNDER $\text{EXACTFRAC} \geq c=0.60$ (5 SEEDS; LiRA $n_{sh}=16$). FEASIBLE UNDER $c > 0$: NONE, GAP-AGG, HARP.
†GAP-INSPIRED AGG. LAPLACE (ANALYTICAL ε ; NOT FULL GAP CODE).
‡SHADOW-FIT MEMGUARD; ϕ = ATTACK IT OPTIMIZES. ¶HARP
FRAC=1 RECOVERS STRONG-LBP ACC-LiRA.

Defense	Feas.	Acc	LiRA	TPR@1%	ϕ	ECE↓	EF
none	yes	0.877	0.813	0.200	0.650	0.051	1.00
GAP-agg ($\sigma=3$, $\varepsilon \approx 41$)†	yes	0.776	0.624	0.039	0.540	0.399	1.00
HARP (release)	yes	0.826	0.674	0.060	0.626	0.033	0.60
LBP equal-mass	no	0.870	0.592	0.009	0.584	0.159	0.00
MemGuard‡	no	0.877	0.667	0.080	0.619	0.063	0.00
LBP $b=0.3$ / HARP-full¶	no	0.707 / 0.715	0.551 / 0.560	0.007 / 0.008	0.565 / 0.572	0.142 / 0.152	0.00

0.674 / 0.060 while keeping Acc 0.826, ECE 0.033, and a 60% replay-stable majority. MemGuard is a credible unconstrained baseline (LiRA 0.667) but ExactFrac=0; equal-mass LBP reaches LiRA 0.592 with ECE 0.159 and ExactFrac=0. Equal-mass can win Acc-LiRA when the SLA is dropped—expected and outside the constrained objective.

b) *Secondary: Acc vs. published strong LBP.*: Against LBP $b=0.3$ [5], HARP raises Acc 0.707 $\rightarrow$ 0.832 at Frac=0.40 because strong noise applies to a minority of nodes (Table II). This is mechanically expected and is not the constrained-feasibility claim.

c) *Membership audit (deployment dial).*: At matched Mass, equal-mass LBP beats HARP on Acc-LiRA (Cora $n_{sh}=16$: Acc 0.870 / LiRA 0.592 vs. 0.826 / 0.674). If ExactFrac is not required, prefer equal-mass LBP. Under the SLA, LiRA guides whether Frac must rise and whether the audit threshold τ is met after CFS.

HARP Mass 325; LBP eq-mass 325; strong LBP 812. MemGuard is a credible unconstrained baseline (LiRA 0.813 $\rightarrow$ 0.667) but ExactFrac-infeasible; HARP is within ≈ 0.01 LiRA of MemGuard with ExactFrac=0.60 and much lower ECE. At

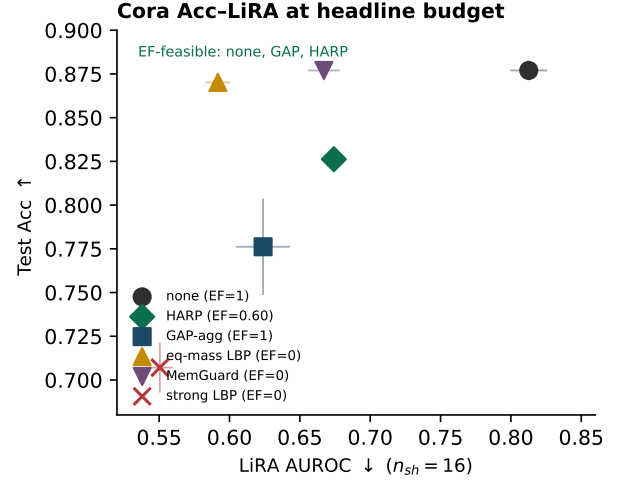


Fig. 4. Cora Acc-LiRA at the headline budget ($n_{sh}=16$, 5 seeds). ExactFrac-feasible: none, GAP-agg, HARP. Equal-mass LBP/MemGuard win unconstrained Acc-LiRA but ExactFrac=0.

$n_{sh}=4$: GTD 0.878/0.538; MaskArmor 0.877/0.580.

GTD/MaskArmor report Acc/LiRA/ECE only (train-time; no Mass/Frac). GAP-agg is ExactFrac-feasible (train-only noise $\Rightarrow$ ExactFrac=1) with finite ε , at Acc/ECE cost. HARP is the high-Acc ExactFrac option without training DP; HARP-full recovers strong-LBP Acc-LiRA under $c=0$. Selective masking keeps ExactFrac=0.60 but raises LiRA.

d) *HARPoGAP: DP trainer as a component.*: Proposition 2 says selective release is post-processing, so composing HARP on a GAP-trained model preserves GAP's (ε, δ). Measured (Cora, 5 seeds, $\sigma=3$): GAP alone Acc 0.770 ± 0.030 , LiRA 0.520; HARPoGAP at Frac=0.40 keeps the same analytical $\varepsilon \approx 41$, ExactFrac=0.60, LiRA 0.517, clean-slice conf AUROC 0.584, at Acc 0.649—composition costs Acc because GAP posteriors already carry training noise. The large ε is the Acc-preserving operating point of our GAP-agg reimplementation ($\sigma=5$ yields $\varepsilon \approx 19$ at Acc 0.44); we do not market $\varepsilon \approx 41$ as a strong privacy guarantee—only as a finite analytical budget that HARP does not inflate. Operators who need a formal ε and selective-release controls can compose; operators who need only ExactFrac keep plain HARP's higher Acc.

Slice ECE localizes the win on Cora (5 seeds): $\text{ECE}_{\text{clean}}=0.018$ vs. $\text{ECE}_{\text{prot}}=0.081$ (population 0.033); clean nodes match undefended calibration, protected nodes carry the noise penalty. The pattern is portable: clean-slice ECE beats undefended population ECE on Cora/Citeseer/Chameleon (0.020/0.083/0.273 vs. 0.053/0.140/0.290), while equal-mass LBP perturbs every response even when its population ECE looks competitive (Citeseer 0.073). ΔAcc vs. strong LBP at Frac=0.40 ranges from +0.040 (Actor) to +0.185 (Photo) across six graphs at matched telemetry ($\text{Mass}_{\text{HARP}}=0.4 \text{ Mass}_{\text{LBP}}$).

e) *Multi-dataset portability.*: Table III checks that selective release remains ExactFrac-feasible across six graphs under

TABLE III

HARP PORTABILITY GRID (GRAPH-SAGE; CORA 5-SEED, OTHERS 3-SEED; $n_{sh}=4$). HEADLINE CORA AUDITS USE $n_{sh}=16$ (TABLE II). TPR = LiRA TPR@1% FPR.

Dataset	Defense	Acc	LiRA	TPR	Mass	Frac
Cora	none	0.877	0.567	0.033	—	—
	LBP $b=0.3$	0.707	0.517	0.019	812	1.00
	HARP	0.832	0.533	0.011	325	0.40
Citeseer	none	0.744	0.591	0.042	—	—
	LBP $b=0.3$	0.620	0.523	0.030	998	1.00
	HARP	0.716	0.548	0.014	399	0.40
Chameleon	none	0.448	0.618	0.059	—	—
	LBP $b=0.3$	0.391	0.539	0.038	683	1.00
	HARP	0.452	0.555	0.011	273	0.40
Actor	none	0.335	0.610	0.037	—	—
	LBP $b=0.3$	0.287	0.549	0.032	2280	1.00
	HARP	0.327	0.561	0.019	912	0.40
PubMed	none	0.876	0.505	0.012	—	—
	LBP $b=0.3$	0.768	0.499	0.010	5915	1.00
	HARP	0.839	0.502	0.012	2366	0.40
Photo	none	0.802	0.500	—	—	—
	LBP $b=0.3$	0.552	0.499	—	2295	1.00
	HARP	0.737	0.501	—	946	0.40

TABLE IV

HARP FRAC SWEEP ON CORA (5 SEEDS; $\sigma_{strong}=0.30$). ACC/MASS ARE BUDGET-INVARIANT; LiRA COLUMN IS $n_{sh}=4$ (SEE TABLE II/ V FOR $n_{sh}=16$).

Frac	Acc $\uparrow$	LiRA $\downarrow$	Mass $\downarrow$
0.20	0.859 [0.855, 0.866]	0.537 [0.532, 0.543]	162
0.30	0.847 [0.843, 0.853]	0.538 [0.533, 0.543]	244
0.40	0.832 [0.826, 0.839]	0.533 [0.525, 0.541]	325
0.60	0.800 [0.793, 0.809]	0.519 [0.512, 0.526]	487
1.00	0.714 [0.699, 0.727]	0.511 [0.503, 0.519]	812

a matched light audit ($n_{sh}=4$; headline Cora uses $n_{sh}=16$). The secondary comparator is published strong LBP ($b=0.3$), which collapses Acc under full-graph noise; HARP recovers 4–19 Acc points at Frac=0.40 while keeping ExactFrac=0.60. This is *not* an unconstrained Acc–LiRA win (equal-mass LBP remains preferred when $c=0$). Heterophilic rows (Chameleon, Actor) show the largest undefended LiRA and the clearest case for a Frac budget; PubMed/Photo are near-chance controls where operators should set Frac=0 unless ExactFrac itself is required.

f) *Frac dial and Mass accounting.*: Table IV sweeps Frac on Cora (5 seeds): Acc falls monotonically as Frac rises under fixed σ_{strong} —the systems Acc–Mass dial. LiRA in this sweep uses $n_{sh}=4$ (cheap dial diagnostics); the headline privacy comparison is Table II at $n_{sh}=16$, and CFS under that budget is Table V.

g) *Constrained Frac Search (matched to the headline audit).*: CFS is only meaningful when τ tracks the attack budget: $\tau=0.55$ is appropriate under $n_{sh}=4$ but infeasible under headline $n_{sh}=16$ (locked HARP LiRA ≈ 0.67). We treat τ as an operator-selected risk tolerance—not a universal privacy target—and run CFS at $n_{sh}=16$ with ExactFrac ≥ 0.60 and study cap $\tau=0.70$ on *validation* Acc/LiRA (Table V); locked Frac=0.40 remains the pre-declared release-only dial reported on the held-out audit. Among Frac $\in \{0, 0.2, 0.3, 0.4\}$, every constructor finds a feasible point; mean selected Frac is

TABLE V

CFS ON CORA AT $n_{sh}=16$ UNDER EXACTFRAC ≥ 0.60 AND LiRA $\leq \tau=0.70$ (3 SEEDS; MEAN SELECTED FRAC/ACC/LiRA). ALL ROWS FEASIBLE_AUDIT=1.

Constructor	Selected Frac	Acc	LiRA
topology (LTE)	0.33	0.838	0.683
ensemble	0.30	0.838	0.681
random	0.23	0.839	0.689

TABLE VI

SESSION BUDGET ON HARP CORA AT $K=20$ (5 SEEDS). $B=1$ FREEZES ACC NEAR ONE-SHOT; $B=20$ MATCHES UNCAPPED. SHARED GLOBAL $B=1$ ACROSS N IDENTITIES BLOCKS SYBIL ACC RECOVERY (RIGHT).

Policy	B	Acc	LiRA	N ids	Per-id $B=1$	Shared $B=1$
Budget reuse	1	0.832	0.540	1	0.825	0.825
Budget reuse	5	0.851	0.540	5	0.874	0.825
Budget reuse	20	0.885	0.545	20	0.876	0.825
Uncapped avg	—	0.886	0.545	—	—	—

0.23–0.33 with Acc ≈ 0.84 —strictly better Acc than the locked Frac=0.40 dial while meeting τ . Tighter $\tau=0.68$ pushes topology to Frac=0.40; looser $\tau=0.75$ selects Frac=0.20. Under ExactFrac alone (no τ), CFS degenerates to Frac=0; the audit cap is what forces a nonzero spend. Locked Frac=0.40 remains the conservative fixed operating point.

h) *Exact scores $\neq$ membership safety.*: Clean nodes are well calibrated (ECE $_{clean}=0.018$), but they are not automatically hard to attack. Under locked HARP, unprotected-slice confidence AUROC is ≈ 0.64 vs. ≈ 0.51 population-wide (Table XI); the mitigation stack below addresses it. Aerni et al. [37] motivate such canary checks (Sec. VI, canaries).

i) *Velocity: multi-query and session budgets.*: Averaging K independent releases denoises Laplace and moves toward the undefended baseline (Cora 5-seed LiRA at $K=1/5/20$: HARP 0.536/0.546/0.551; $n_{sh}=4$). A session policy of at most B fresh draws per *authenticated principal*—reusing the last draw thereafter—caps this drift (Table VI). **Enforcement assumption:** server-side state bound to authenticated identities with rate limits. Sybil-capable attackers who mint many principals can bypass per-identity B ; Table VI shows $N=5$ identities under per-id $B=1$ recover Acc ≈ 0.874 (near uncapped), while a *shared global* $B=1$ pool keeps Acc at the one-shot floor (0.825). Deploy shared B (or authenticated single-principal sessions) when Sybil velocity is in scope; Frac and B remain complementary controls.

j) *Constructors: a privacy effect, not just an interface.*: The decisive constructor metric is *clean-slice* conf AUROC: ranking removes vulnerable members from the bit-exact slice (Table VII). At matched ExactFrac=0.60 (5-seed Cora), slice AUROC falls with ranking quality: random 0.669 ± 0.015 , topology 0.647 ± 0.012 , audit 0.616 ± 0.016 , ensemble 0.594 ± 0.010 . Paired tests vs. random: audit $\Delta=0.053$ ($p<10^{-3}$), ensemble $\Delta=0.075$ ($p<10^{-3}$); Acc is equal-or-better (ensemble 0.813 vs. random 0.811). At the headline budget ($n_{sh}=16$, Frac=0.40), ensemble also lowers population LiRA vs. topology (0.646 vs. 0.674)—constructors are pri-

TABLE VII

RELEASE-ONLY CONSTRUCTORS AT FRAC=0.40 (CORa, 5 SEEDS; MEAN $\pm$ 1.96 SE). SLICE = CLEAN-SLICE CONF AUROC. LiRA HERE $n_{sh}=4$; HEADLINE IS TABLE II.

Policy	Feas.	Acc	LiRA	Slice $\downarrow$	EF
none	yes	0.877 $\pm$ 0.005	0.567 $\pm$ 0.006	—	1.00
GAP-agg ($\sigma=3$)	yes	0.776 $\pm$ 0.024	0.518 $\pm$ 0.008	—	1.00
LBP equal-mass	no	0.870 $\pm$ 0.005	0.520 $\pm$ 0.011	—	0.00
LBP $b=0.3$ / HARP-full	no	0.707 / 0.715	0.517 / 0.512	—	0.00
sel. audit	yes	0.820 $\pm$ 0.010	0.526 $\pm$ 0.009	0.616 $\pm$ 0.016	0.60
sel. ensemble	yes	0.817 $\pm$ 0.011	0.527 $\pm$ 0.014	0.594$\pm$0.010	0.60
sel. random	yes	0.810 $\pm$ 0.006	0.526 $\pm$ 0.007	0.669 $\pm$ 0.015	0.60
sel. entropy	yes	0.805 $\pm$ 0.010	0.530 $\pm$ 0.014	—	0.60
sel. topology (LTE)	yes	0.826 $\pm$ 0.006	0.536 $\pm$ 0.008	0.647 $\pm$ 0.012	0.60

TABLE VIII

RAWEXACTFRAC VS REPLAYFRAC (CORa, 3 SEEDS). GLOBAL-CACHE AND SEEDED LBP RESTORE REPLAY/CACHE WITHOUT RAW POSTERIOR EQUALITY; HARP MEETS EXACTFRAC= c WITHOUT A NON-RAW LEDGER.

Policy	RawEF $\uparrow$	Replay $\uparrow$	Cross $\uparrow$	Cache $\uparrow$
none	1.00	1.00	1.00	0.87
LBP fresh	0.00	0.00	0.00	0.00
LBP sticky ($B=1$)	0.00	1.00*	0.00	0.62
LBP global cache	0.00	1.00	1.00	0.87
LBP seeded (node,epoch)	0.00	1.00	1.00	0.87
HARP (Frac 0.40)	0.60	0.60	0.60	0.77

*Within-client sticky replay; cross-client ExactFrac stays 0. Clean-slice threshold flicker $\approx$ 0.09 for fresh/sticky LBP and 0 for global/seeded/HARP.

TABLE IX

IN-PROCESS RESPONSE-CACHE MICROBENCHMARK (CORa; 20K ZIPF QUERIES; CONTENT KEYS ON RELEASED VECTORS). ABSOLUTE QPS/LATENCY ARE NOT END-TO-END API NUMBERS.

Workload	Policy	Hit	QPS	p50 ms	p95 ms
zipf	none	0.937	4.0×10^6	<0.001	<0.001
zipf	lbp_fresh	0.000	9.3×10^3	0.104	0.116
zipf	harp_fresh	0.606	6.2×10^5	<0.001	0.004
zipf	lbp_B1	0.937	1.5×10^5	<0.001	0.101
zipf	harp_B1	0.937	2.2×10^6	<0.001	<0.001

marily a slice dial, with a measurable population effect under strong audits. Hop $k \in \{0, 1, 2\}$ at matched Frac changes Acc by $\lesssim 0.01$ (release-only); selectivity, not train-time alignment, carries ExactFrac.

k) *Why ExactFrac matters (RawExactFrac $\neq$ ReplayFrac).*: Caches and ledgers can stabilize previously sampled noisy outputs [40], [41]; that is ReplayFrac, not ExactFrac. Table VIII (Cora; 3 seeds; 8 clients; 20k Zipf; shared LRU 1024) makes the separation concrete: fresh LBP has both fractions ≈ 0 ; per-client sticky LBP raises within-client ReplayFrac but not cross-client equality or ExactFrac; *global* first-draw caches and deterministic seeded LBP achieve ReplayFrac=1 and high cache hits with ExactFrac=0; HARP alone among noisy policies meets ExactFrac=0.60 without storing non-raw scores. Table X confirms clean-slice bit-stability; Table IX is an *in-process response-cache microbenchmark* (not end-to-end networked API throughput; absolute QPS/latency are mechanism artifacts on Cora).

TABLE X

MEASURED (NOT DEFINITIONAL) REPLAY BEHAVIOR ACROSS INDEPENDENT RE-QUERIES (CORa, 3 SEEDS): BITWISE-EQUAL FRACTION, TOP-1 LABEL FLIPS, AND THRESHOLD ($p_{top} > 0.5$) ROUTE FLIPS. HARP'S FLIPS ARE CONFINED TO THE PROTECTED MINORITY; THE CLEAN 60% IS BIT-STABLE WITH ZERO FLICKER.

Policy	Bit-equal re-query	Top-1 flip	Thresh. flip
none	1.000	0.000	0.000
GAP-agg	1.000	0.000	0.000
LBP eq-mass	0.000	0.027	0.101
HARP (Frac 0.40)	0.600	0.146*	0.181*

*Entirely within the protected 40% ($\sigma=0.30$); clean-slice flip rate is exactly 0. LBP spreads lower-amplitude flicker over *all* nodes, so no response is replay-stable.

TABLE XI

LTE-AWARE ADAPTIVE QUERIES ON CORa HARP (5 SEEDS; CONF AUROC IS ATTACK-BUDGET FREE; LiRA@0.4 USES $n_{sh}=4$).

Query set	Conf@0.4	Conf@0.6	LiRA@0.4
Population	0.518	0.473	0.533
Unprotected (clean)	0.645	0.633	0.542
Boundary (clean)	0.634	0.628	0.538
Top-decile LTE	0.641	0.511	0.529

l) *Clean-slice residual risk (first-class) and the mitigation stack.*: Unprotected nodes are *not* claimed safe (Table XI); clean-slice conf AUROC rises to 0.63–0.67 while population LiRA stays ≈ 0.53 . HARP attacks this from three composable directions (all at ExactFrac=0.60, 5 seeds): (i) *constructor*: ensemble ranking removes vulnerable members from the exact slice, $0.669 \rightarrow 0.594$ (Table VII); (ii) *deterministic confidence smoothing (DCS)*: released rows on the clean slice with $p_{max} > \theta$ are flattened by a fixed temperature ($\theta=0.9$, $T=3$)—a deterministic, argmax-preserving post-map, so re-queries stay bitwise stable, caches and audit replay still work, and routing decisions are unchanged; slice AUROC falls $0.650 \rightarrow 0.618$ under topology ranking and $0.591 \rightarrow 0.576$ on top of ensemble, with *zero* Acc change; (iii) *slice-aware CFS*: adds a slice-AUROC cap to the search. DCS trades raw-score fidelity on smoothed rows for slice privacy while preserving ReplayFrac and threshold stability; when ExactFrac (raw $\hat{p}_v$) is contractual, skip DCS. The stacked floor (≈ 0.575) is structural at $c=0.60$: a 60% deterministic slice must retain some membership signal. Lower c , or use GAP-agg / HARP $\circ$ GAP, if tighter slice control is required.

m) *Inductive edge exclusion.*: Dropping test-incident edges (Cora, 5 seeds) raises undefended LiRA to 0.818; HARP recovers Acc $0.476 \rightarrow 0.601$ vs. strong LBP at Frac=0.40.

C. When to spend: volume $\times$ structure

Table XII(a): undefended LiRA concentrates in small/mid heterophilic cells; Photo/PubMed/ogbn-arxiv sit near chance under matched shadows—audit first, then set Frac. Table XII(b) reports the honest shadow-count story on Cora for none / LBP / HARP. Absolute LiRA rises with $n_{shadows}$ for every defense [4]; Acc is invariant. HARP does *not* beat strong LBP on LiRA at large shadow counts (Cora $n_{sh}=64$: LBP 0.561 vs. HARP 0.677)—expected when ExactFrac=0

TABLE XII

AUDIT CONTROLS (GRAPHSAGE; 3 SEEDS). *HETEROPHILIC.
(B) SHADOW SCALING (NOT HARP-ONLY). ACC INVARIANT IN (B): NONE
0.879, LBP 0.716, HARP 0.836.

(a) Undef. LiRA ($n_{sh}=4$)				(b) Cora LiRA / TPR@1%			
Dataset	n	Acc	LiRA	n_{sh}	none	LBP	HARP
Chameleon*	2277	0.449	0.623	4	0.568/0.033	0.517/0.022	0.534/0.014
Cora	2708	0.879	0.568	16	0.808/0.201	0.555/0.008	0.670/0.057
Actor*	7600	0.335	0.610	64	0.847/0.265	0.561/0.008	0.677/0.075
Photo	7650	0.825	0.500				
PubMed	19717	0.876	0.505				
ogbn-arxiv	169343	0.666	0.503				

is allowed. What HARP keeps is (i) strictly lower LiRA than undefended and (ii) Acc recovery vs. strong LBP ($\Delta\text{Acc} +0.12$, invariant in n_{sh}). TPR@1% FPR follows the same pattern.

D. Canaries and large-graph volume

Cora canary checks: HARP suppresses planted/top-decile confidence cues relative to none/LBP (top-decile conf AUROC 0.551/0.548/0.266 for none/LBP/HARP; planted 0.527/0.543/0.423); population LiRA remains primary. Table XIV: on ogbn-arxiv ($\sim 169k$; NeighborLoader; gate off, warmup 3) [28], strong LBP drops Acc to 0.179 [0.175, 0.184]. HARP recovers 0.599 [0.597, 0.601] (paired $\Delta\text{Acc} +0.420$ [0.414, 0.425]; 3 seeds) at $\text{Frac}=0.40$. Undefended LiRA stays near chance at $n_{sh} \in \{2, 4\}$ and also at $n_{sh}=16$ (LiRA 0.510; Acc 0.666)—so the Acc recovery is a serving result under an audit-null regime, not under-powered shadows. On ogbn-products ($\sim 2.4M$ nodes) [28], Table XIII reports a serving probe plus BFS LiRA at $n \in \{15k, 40k\}$ and $n_{sh} \in \{2, 4, 8\}$. Full-graph products LiRA OOMs on this host; we do *not* claim a million-node privacy audit. Across standard-recipe probes LiRA stays near chance while HARP recovers Acc vs. strong LBP (e.g., $15k/n_{sh}=8$: HARP Acc 0.693 vs. LBP 0.343).

a) *Canary-stressed probe (above-chance signal at scale)*.: Standard products/arxiv recipes sit near chance, so Acc recovery alone is a weak privacy claim. We plant 300 unique-feature canaries on the 15k BFS probe (half in train / half held out; $n_{sh}=4$; 3 seeds): undefended canary-LiRA is ≈ 1.00 while population LiRA is 0.593 (Table XIII). HARP cuts canary-LiRA to 0.891 (still high) at Acc 0.685 with $\text{ExactFrac}=0.60$; strong LBP reaches 0.805 only by collapsing Acc to 0.379 at $\text{ExactFrac}=0$. Population LiRA barely moves (≈ 0.59) under either release defense—defense-aware shadows recalibrate—so the scale takeaway is Acc-preserving ExactFrac under a hard canary stress, not a population-LiRA win at 15k. We do *not* claim a full 2.4M-node privacy audit.

If audit LiRA ≈ 0.5 , prefer $\text{Frac}=0$; if LiRA $\gg 0.5$ or $\text{ExactFrac} \geq c$ is required, set Frac on val Acc.

b) *Operator workflow*.: Audit with defense-aware LiRA ($n_{sh} \geq 16$ when affordable); if LiRA ≈ 0.5 set $\text{Frac}=0$; else CFS on val Acc under (c, τ) (Table IV); prefer ensemble constructors + optional DCS for slice risk; enforce shared $B=1$ under Sybil velocity; use HARPoGAP when a legal ε is required.

TABLE XIII

OGBN-PRODUCTS: STANDARD BFS LADDER VS. CANARY-STRESSED 15K
($n_{sh}=4$, 3 SEEDS). CAN. = CANARY LiRA.

Setting	Acc	LiRA	Can.	EF
Full load none (2.45M)	0.695	—	—	1.00
Std. 15k none/LBP/HARP	0.876/0.343/0.693	0.528/0.501/0.513	—	1/0/0.60
Canary none	0.861	0.593	0.997	1.00
Canary LBP $b=0.3$	0.379	0.581	0.805	0.00
Canary HARP	0.685	0.592	0.891	0.60

TABLE XIV

OGBN-ARXIV VOLUME (3 SEEDS). LiRA_{2/4} ARE DEFENSE-AWARE;
UNDEFENDED LiRA AT $n_{sh}=16$ IS 0.510 (NEAR-CHANCE CREDIBILITY
CHECK).

Def.	Acc	LiRA ₂	LiRA ₄	Mass
none	0.666	0.502	0.503	—
LBP	0.179	0.498	0.501	50 803
HARP	0.599	0.514	0.505	20 322

VII. DISCUSSION

HARP is a Big-Data score-serving mechanism for $\text{ExactFrac} \geq c > 0$ —not a claim to beat uniform noise when $c=0$, and not an equation of ReplayFrac with raw-posterior equality. The venue fit is prediction-API infrastructure under Volume/Velocity/Veracity: selective release, cache/replay accounting, and audits on high-volume graphs—membership privacy enters as an operator dial on that serving stack. ExactFrac formalizes a raw-score SLA that generic response caches do not automatically provide [40], [41]; $c=0.60$ is a study default, and sticky/global/seeded noise are not substitutes (Table VIII). GAP-agg already meets $\text{ExactFrac}=1$ with finite (often large) ε ; HARP’s complementary claim is high-Acc ExactFrac without training DP, plus constructors/CFS/slice controls (Prop. 2). **Limits:** (i) $\text{ExactFrac} > 0 \Rightarrow$ infinite global ε on released scores; (ii) clean-slice AUROC stays above chance at $c=0.60$ —structural; (iii) CFS τ is an operator dial that must track the LiRA budget; (iv) products privacy is a BFS/canary ladder; (v) GAP-agg/MemGuard are reimplementations; (vi) serving benches are in-process microbenchmarks, not production traces. Future: learned constructors, distributed full-graph LiRA, production API traces.

VIII. CONCLUSION

HARP equips GNN prediction APIs with ExactFrac -constrained score serving: selective release meets raw-posterior $c > 0$ where fresh and stateful-noisy uniform policies cannot, constructors cut clean-slice attack AUROC, CFS configures Frac under operator (c, τ) , and HARPoGAP preserves training DP for public-metadata constructors. Headline evidence uses $n_{sh}=16$; residual clean-slice risk and scoped large-graph Acc/serving probes are reported honestly. Use training-time DP (alone or under HARP) when a legal global ε is required.

a) *Reproducibility*.: Code (defenses/harp.py, run_nsh16_headline.py, run_cfs_nsh16.py, run_exactfrac_sla_evidence.py,

run_products_canary_leakage.py, run_bulletproof.py), locked configs, and frozen CSVs: <https://github.com/krishharma/privacy-gnn>. Headline LiRA $n_{\text{shadows}}=16$; CFS uses $\tau=0.70$ on validation; RawExactFrac SLA evidence in Table VIII.

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